

Total No. of Questions : 10]

SEAT No. :

P2464

[Total No. of Pages : 2

[5253] - 187

T.E. (Computer Engineering)

EMBEDDED OPERATING SYSTEMS

(2012 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer: Q.No. 1 or Q.No. 2, Q.No. 3 or Q.No. 4, Q.No. 5 or Q.No. 6, Q.No. 7 or Q.No. 8, Q.No. 9 or Q.No. 10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a) What is priority inversion? What are the solutions available to handle the priority inversion? [6]
- b) What is Readers-Writers problem? [4]

OR

- Q2) a) Explain the RISC architectural features of ARM. [4]
- b) What are the quality points that rate a scheduling algorithm? [4]
- c) Name four Embedded Operating Systems. [2]

- Q3) a) What are the reasons for the growth and popularity of Embedded Linux? [3]
- b) Explain steps involved in compiling Linux Kernel for ARM-XScale architecture as a target. [5]
- c) Name four executables or binaries of Embedded Linux. [2]

OR

- Q4) a) What is cross development environment for Linux? Elaborate. [5]
- b) What is Busy Box? How to configure it? Explain its usefulness in embedded systems. [5]

- Q5) a) Name and explain the typical bootloader used for embedded/target board? Also mention the commands available with such a bootloader. [6]
- b) How to format and partition a USB stick? Explain the commands used. [7]
- c) How flash memory is used in embedded/target board? What are its limitations? [4]

OR

P.T.O.

- Q6)** a) What are the different types of device drivers? Explain depmod and rmmod. [6]
b) How MTD utility is useful for target boards? How to enable and use MTD services? [6]
c) What is Das U-Boot? What are U-Boot command sets? [5]

- Q7)** a) What is core dump? How to debug a core dump? [7]
b) Recognize and explain the following: [6]
i) ps
ii) strace
iii) mtrace
c) What is KGDB? [4]

OR

- Q8)** a) What is a stepper motor? How to interface BBB with Stepper motor?[7]
b) How do modern processors and compilers make it difficult to debug Linux kernel? [6]
c) What are the challenges faced while debugging Linux application code?[4]

- Q9)** a) Explain in details steps involved while porting Linux on a target board.[8]
b) What are the scheduling policies used by Linux to schedule real time processes? [6]
c) Explain Zygote for Embedded Android. [2]

OR

- Q10)**a) What are real-time processes? Which latency periods affect their performance? [8]
b) What is real-time scheduling in Linux? [4]
c) Explain System Server and Activity Manager for Embedded Android.[4]

